

Who Makes Unusual Combinations?

A Scientist-Level View of Scientific Recombination

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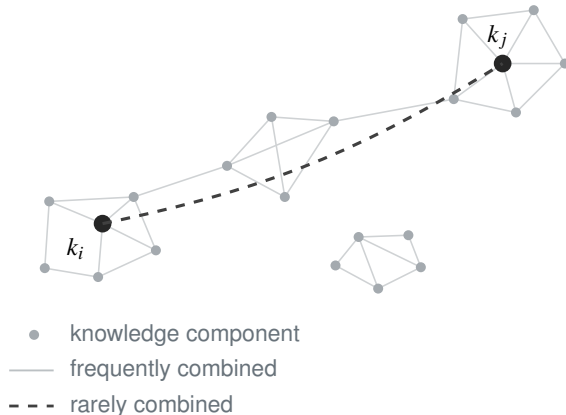
Seoul National University

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Unusual combination

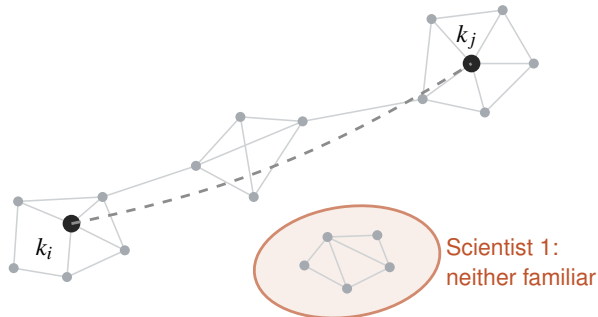
Unusual combinations matter, as they are associated with novelty and impact.



Conceptual illustration; our analyses do not use network structure.

Unusual combination—who makes it?

Yet **combinations do not assemble themselves**; they are made by (different) scientists.

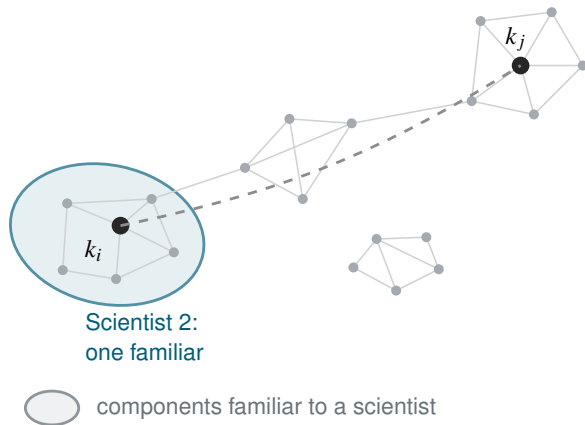


 components familiar to a scientist

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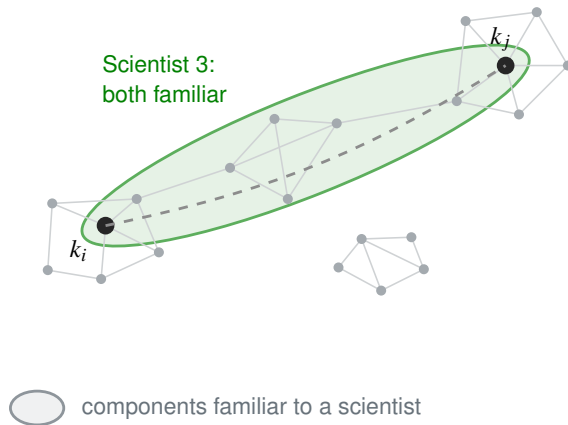
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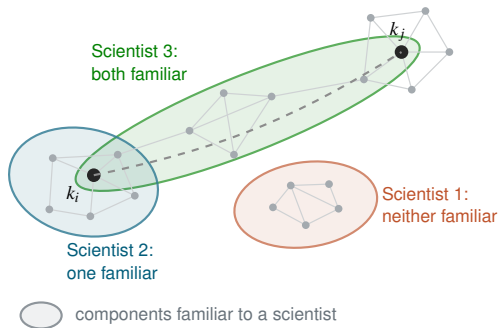
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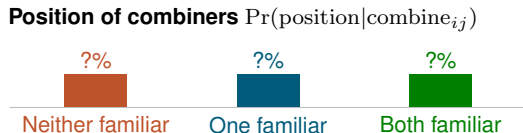
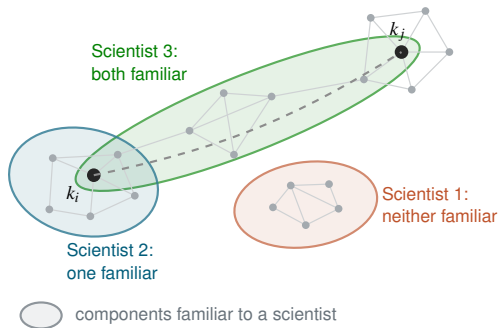
Unusual combination—who makes it?

- Scientists are in *different positions* relative to the same unusual combination.



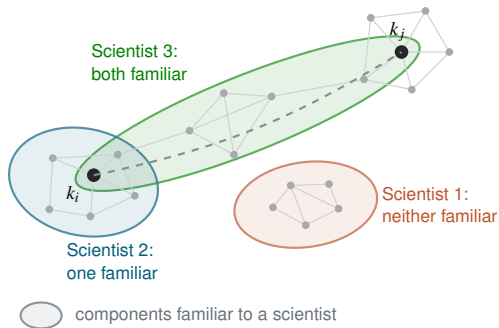
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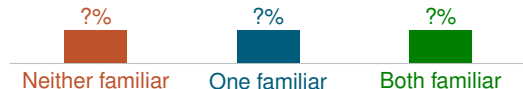


Unusual combination—who makes it?

- Scientists are in *different positions* relative to the same unusual combination.
- From *which position* are unusual combinations made?
- How common is *each position* among scientists?



Position of combiners $\Pr(\text{position}|\text{combine}_{ij})$

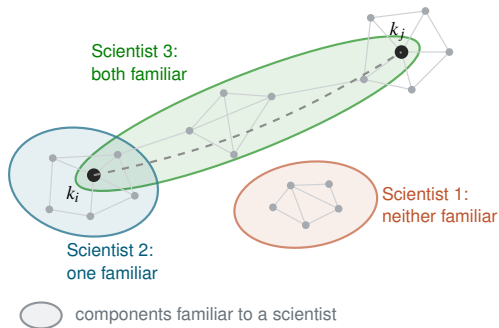


Distribution of positions $\Pr(\text{position})$

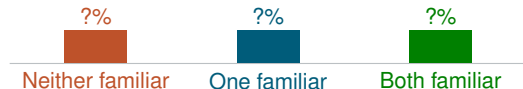


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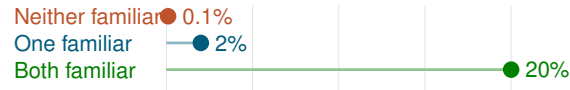
Position of combiners $\Pr(\text{position}|\text{combine}_{ij})$



Distribution of positions $\Pr(\text{position})$



Probability of combining $\Pr(\text{combine}_{ij}|\text{position})$



Research questions

For a combination of two components (k_i, k_j) , we ask:

RQ1: Distribution of positions

Where are scientists positioned relative to each component?

RQ2: Position of combiners

From which positions are combinations made?

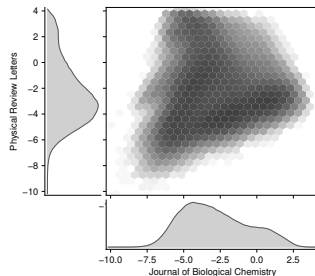
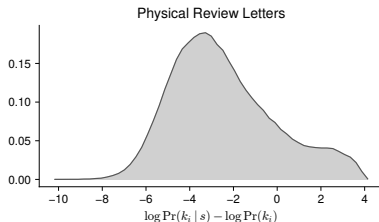
RQ1: Distribution of positions

Where are scientists positioned relative to each component?

- For each scientist s , we estimate their relative propensities of using a component k_i :

$$\tilde{p}_{i|s} = \log \frac{\Pr(k_i | s)}{\Pr(k_i)}$$

- For a combination of two components (k_i, k_j) , we examine the distribution of scientists' positions $(\tilde{p}_{i|s}, \tilde{p}_{j|s})$ for all scientists $s \in \mathcal{S}$.

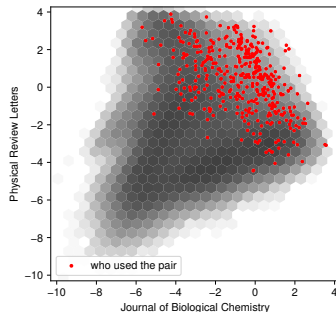


Research questions

RQ2: Position of combiners

From which positions are combinations made?

- For each pair (k_i, k_j) , we examine the distribution of positions $(\tilde{p}_{i|s}, \tilde{p}_{j|s})$ among scientists who actually make the combination.

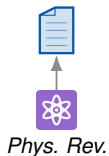


Data and Methods

- Scientific publication data (SciSciNet-v2, OpenAlex).

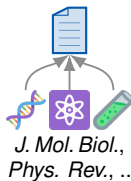
Empirical setting

- Scientific publication data (SciSciNet-v2, OpenAlex).
- **Component** k_i : a journal cited in a paper's references.



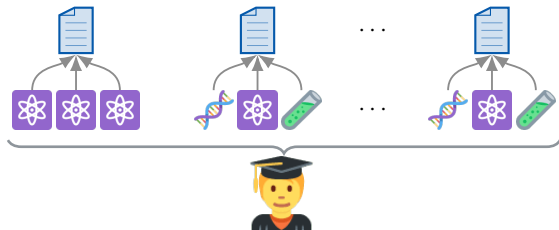
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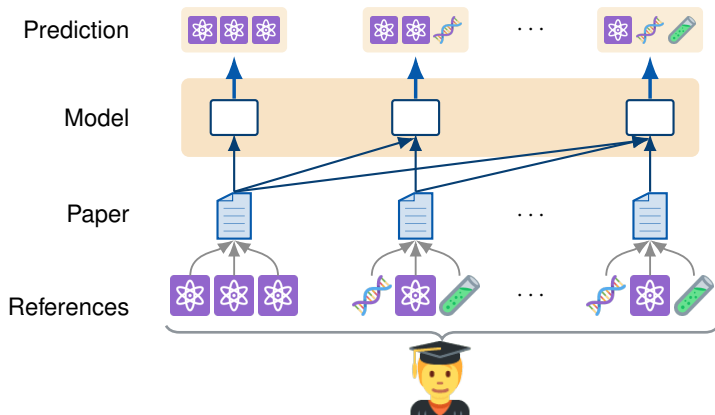
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- **Research trajectory** R_s : a scientist s 's first-authored papers.



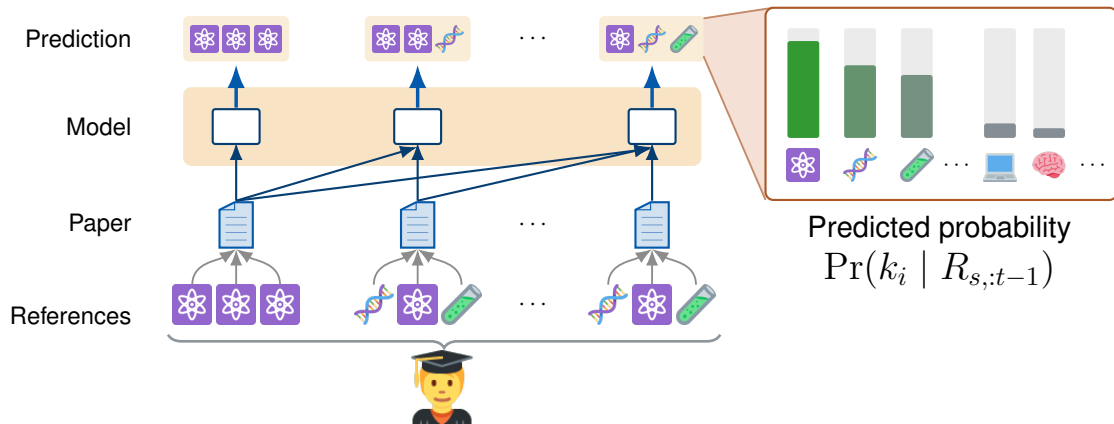
Deep sequential model of scientists' trajectories

- A hierarchical transformer model is trained to predict which journals a scientist will cite in their next paper, given their previous trajectory $R_{s,t-1}$.



Deep sequential model of scientists' trajectories

- A hierarchical transformer model is trained to predict which journals a scientist will cite in their next paper, given their previous trajectory $R_{s,:t-1}$.
 - The model estimates scientists' propensity for citing k_i at their next paper.

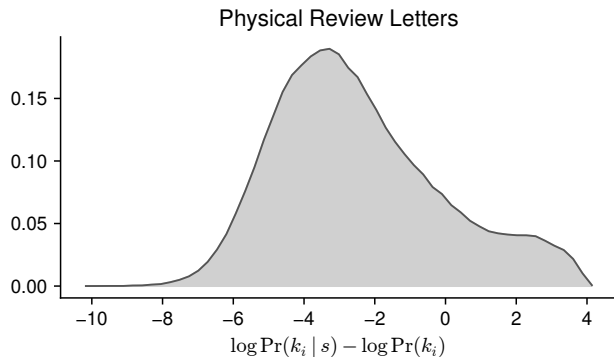


- We use **2012** as an illustrative focal year.
- We train the model to predict which journals a scientist will cite in their papers published in 2002–2011, given their prior research trajectory (1970–2010).
- We score scientists' positions based on their prior research trajectories in 1970–2011.
 - 389,232 eligible scientists.
- We observe how scientists co-cited journals in their papers published in 2012.

Distribution of Positions

Distribution of propensities to use components

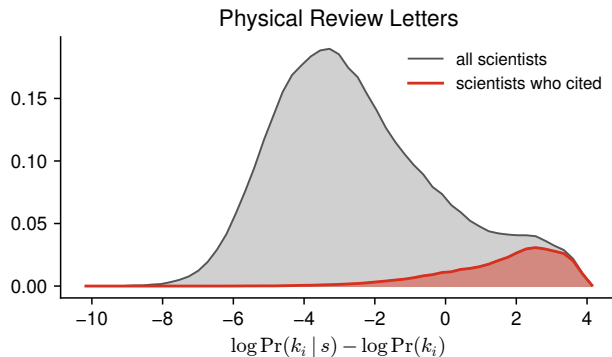
- We score the propensities of using components, $\tilde{p}_{i|s}$, for all scientists $s \in \mathcal{S}$ and components $k_i \in \mathcal{V}$.
 - $\tilde{p}_{i|s} > 0$ indicates that scientist s is more reachable to component k_i than a random scientist.



Distribution of $\tilde{p}_{i|s}$ for component *Physical Review Letters* for all scientists $s \in \mathcal{S}$.

Distribution of propensities to use components

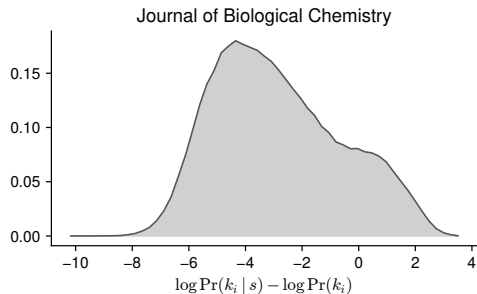
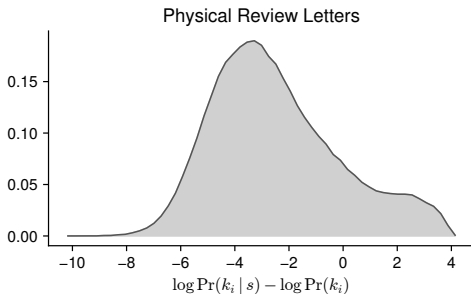
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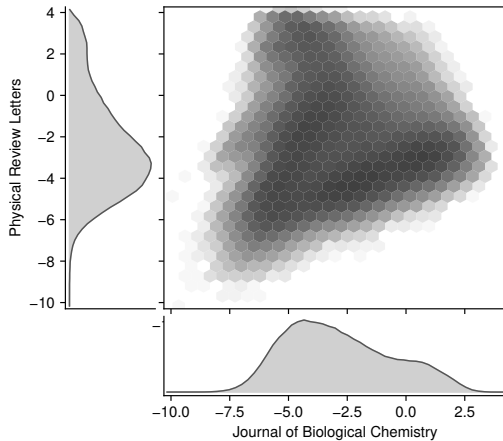
Joint distribution of propensities to use two components

- Consider a pair of *Physical Review Letters* and *Journal of Biological Chemistry*.
- Who are likely to reach both components?



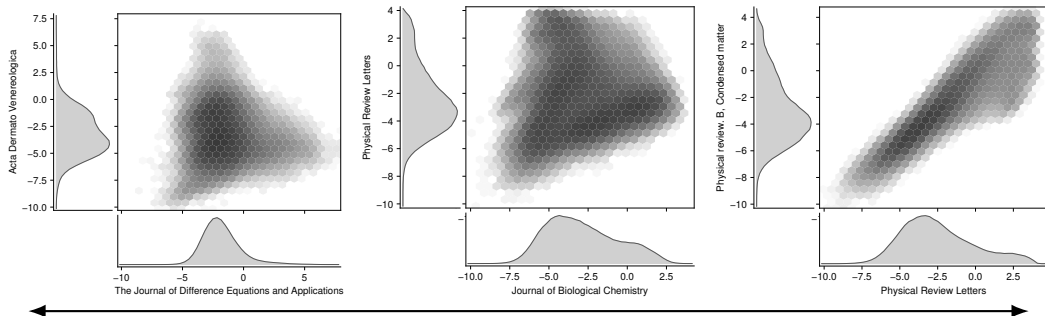
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How propensity distributions of two components overlap?

- We compare the distributions of positions for different component pairs.
- How often are both propensities $\tilde{p}_{i|s}$ and $\tilde{p}_{j|s}$ high for the same scientists s ?



Segregation

Each component tends to be reachable by different scientists

$$\mathbb{E}_s[p_{i|s} \cdot p_{j|s}] < \mathbb{E}_s[p_{i|s}] \cdot \mathbb{E}_s[p_{j|s}]$$

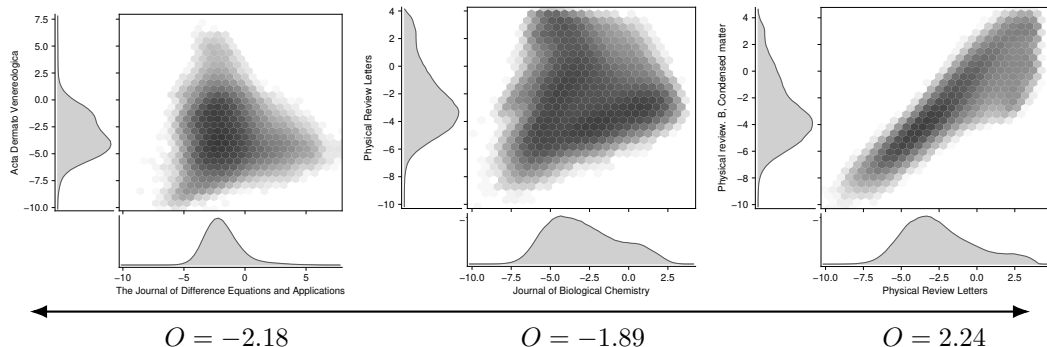
Overlap

Both components tend to be reachable by the same scientists

$$\mathbb{E}_s[p_{i|s} \cdot p_{j|s}] > \mathbb{E}_s[p_{i|s}] \cdot \mathbb{E}_s[p_{j|s}]$$

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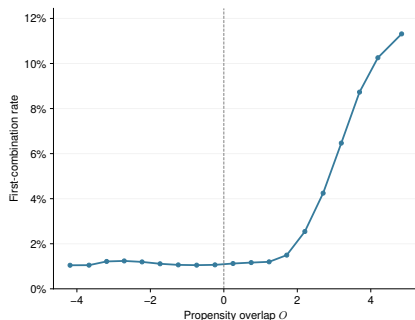
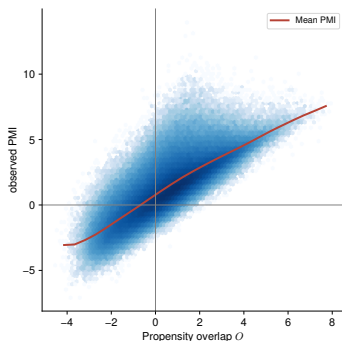
Propensity overlap $O_{ij} = \log \frac{\mathbb{E}_s[p_{i|s} \cdot p_{j|s}]}{\mathbb{E}_s[p_{i|s}] \cdot \mathbb{E}_s[p_{j|s}]}$

Propensity overlap and unusual combinations

- When O_{ij} is high, components k_i, k_j are more likely to:
 - be combined more often than expected, *i.e.*, have higher PMI (**typical combinations**).

$$\text{PMI}_{ij} = \log \frac{\Pr(k_i, k_j)}{\Pr(k_i) \Pr(k_j)}$$

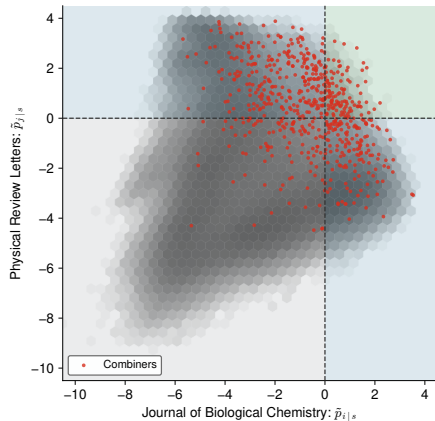
- be combined for the first time if they have not been combined before.



Position of Combiners

Position of combiners

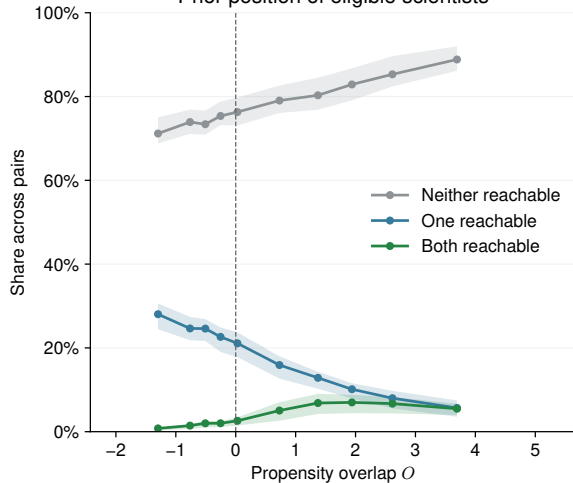
- Where are the actual combiners positioned?



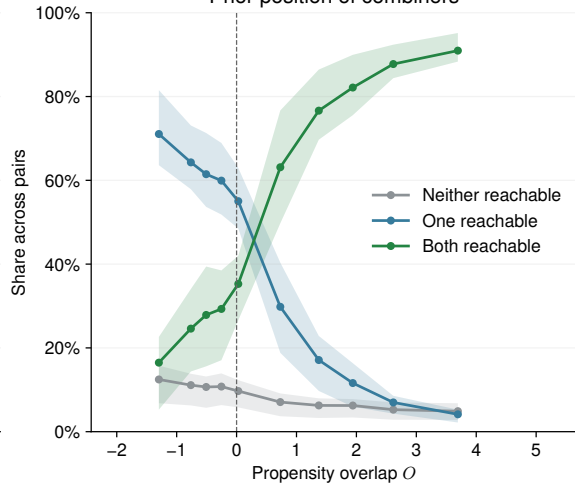
- Among 389,232 eligible scientists,
 - Neither reachable: 276,597 (71.06%)
 - One reachable: 111,021 (28.52%)
 - Both reachable: 1,614 (0.41%)
- Among 629 combiners of the pair,
 - Neither reachable: 102 (16.22%)
 - One reachable: 417 (66.30%)
 - Both reachable: 110 (17.49%)

Position of combiners across component pairs

Prior position of eligible scientists

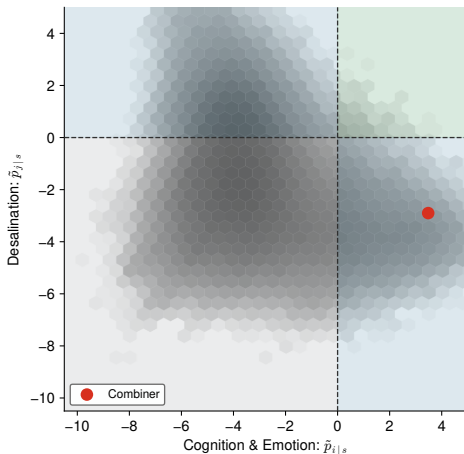


Prior position of combiners



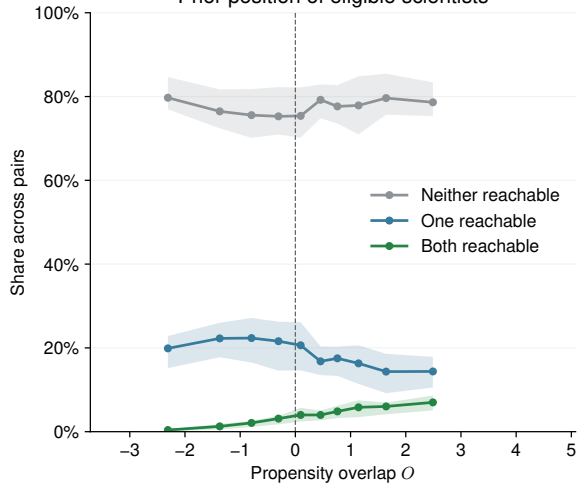
Position of the first combiners

- Where are the first combiners positioned, for pairs that have not been combined before?

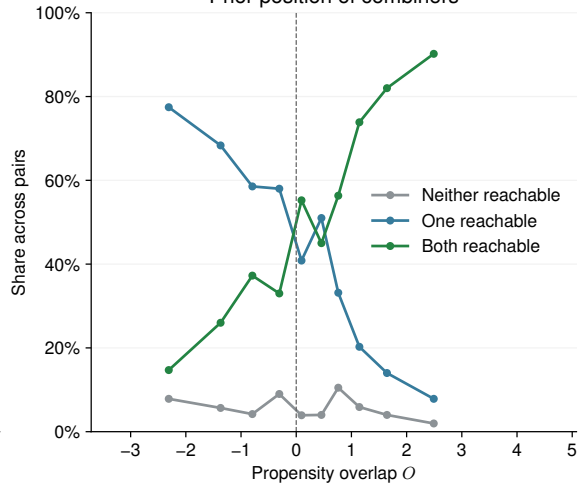


Position of the first combiners across combined pairs

Prior position of eligible scientists



Prior position of combiners



A scientist-level view of scientific recombination

● Summary

- Combinations that are unusual to science may be usual to some scientists.
- Both-reachable scientists can be rare in the population, but they are more likely to make the combination.
- Unusual pairs are more frequently made by one-reachable scientists, as they are much more common in the population.

● Next steps

- How does a pair's combiner composition **evolve** over time?
- For the same combination, does paper **impact** vary with the combiner's prior position?

● Limitations

- Journal-based knowledge representation.
- Complementary knowledge within teams.